

1) Problem 7.1

Disclaimer: In the homework file uploaded, question states "nonconducting" medium. However, in the textbook question is stated as "conducting" medium. I will solve the question for both cases separately

a)

→ For "nonconducting" medium, general Maxwell equations are:

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}, \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \nabla \cdot \vec{B} = 0, \quad \nabla \cdot \vec{D} = \rho_v$$

If medium is free of sources, with parameters ϵ, μ

$$\nabla \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}, \quad \nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}, \quad \nabla \cdot \vec{H} = 0, \quad \nabla \cdot \vec{E} = 0$$

Taking the divergence of both sides of first equation:

$$\nabla \times (\nabla \times \vec{H}) = \underbrace{\nabla(\nabla \cdot \vec{H})}_0 - \nabla^2 \vec{H} = \nabla \times \left(\epsilon \frac{\partial \vec{E}}{\partial t} \right) = \epsilon \frac{\partial}{\partial t} (\nabla \times \vec{E}) = \epsilon \frac{\partial}{\partial t} \left(-\mu \frac{\partial \vec{H}}{\partial t} \right)$$

$$\boxed{\nabla^2 \vec{H} - \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2} = 0} \Rightarrow \text{Wave Equation governing } \vec{H}$$

→ Similarly for $\vec{E}$

$$\nabla \times (\nabla \times \vec{E}) = \underbrace{\nabla(\nabla \cdot \vec{E})}_0 - \nabla^2 \vec{E} = -\mu \frac{\partial}{\partial t} (\nabla \times \vec{H}) = -\mu \frac{\partial}{\partial t} \left(\epsilon \frac{\partial \vec{E}}{\partial t} \right)$$

$$\boxed{\nabla^2 \vec{E} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0} \Rightarrow \text{Wave Eq. governing } \vec{E}$$

If medium is conductive $\Rightarrow$ (still source free with μ, ϵ, σ)

$$\nabla \times \vec{H} = \sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t}, \quad \nabla \cdot \vec{H} = 0$$

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}, \quad \nabla \cdot \vec{E} = 0$$

$$\nabla \times (\nabla \times \vec{H}) = -\nabla^2 \vec{H} = \sigma (\nabla \times \vec{E}) + \epsilon \frac{\partial}{\partial t} (\nabla \times \vec{E}) = -\sigma \mu \frac{\partial \vec{H}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2} \Rightarrow \boxed{\nabla^2 \vec{H} - \sigma \mu \frac{\partial \vec{H}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2} = 0}$$

Similarly for $\vec{E}$

$$\nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E} = -\mu \frac{\partial}{\partial t} \left(\sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t} \right) = -\sigma \mu \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} \Rightarrow \boxed{\nabla^2 \vec{E} - \sigma \mu \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0}$$

b) For time-harmonic field we can express $\vec{E}, \vec{B}$ by vector phasors if they vary sinusoidally with time.

In phasor domain multiplying with $j\omega$ is equivalent to time derivative
Then,

$$\nabla^2 \vec{H}_s - \mu \epsilon (j\omega)^2 \vec{H}_s = 0 \Rightarrow \boxed{\nabla^2 \vec{H}_s + \mu \epsilon \omega^2 \vec{H}_s = 0}$$

$$\left(\frac{\partial}{\partial t}\right) \rightarrow (j\omega)$$

Replacing $\frac{\omega^2}{4\pi^2} = \frac{\omega^2}{\mu \epsilon} = k^2 = \mu \epsilon \omega^2$

$$\boxed{\nabla^2 \vec{H}_s + k^2 \vec{H}_s = 0}$$

where $\vec{H}_s$ is the phasor of $\vec{H}$

Similarly for $\vec{E}$

$$\nabla^2 \vec{E}_s - \mu \epsilon (j\omega)^2 \vec{E}_s = 0 \Rightarrow$$

$$\boxed{\begin{aligned} \nabla^2 \vec{E}_s + \mu \epsilon \omega^2 \vec{E}_s &= 0 \\ \nabla^2 \vec{E}_s + k^2 \vec{E}_s &= 0 \end{aligned}}$$

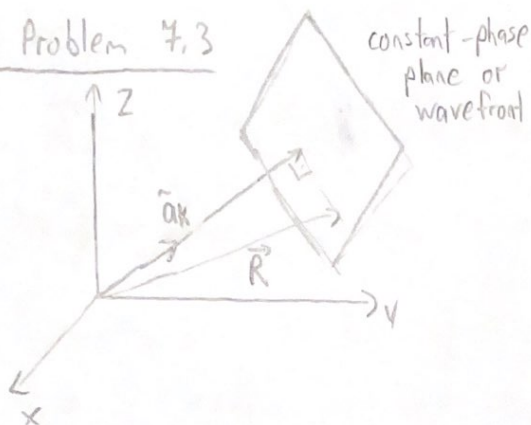
where $k = \omega \sqrt{\mu \epsilon}$

For "conducting" medium, equations transform into

$$\boxed{\nabla^2 \vec{H}_s + k^2 \vec{H}_s = j\omega \mu \sigma \vec{H}_s}$$

$$\boxed{\nabla^2 \vec{E}_s + k^2 \vec{E}_s = j\omega \mu \sigma \vec{E}_s}$$

2) Problem 7.3



Assume we have a wave in that moves in an arbitrary direction $\hat{a}_k$ and also $\vec{H}_0$ be the vector indicating the amplitude and direction of the wave normal to $\hat{a}_k$ (or located in constant-phase plane). $\Rightarrow \vec{H}_0 = H_0 \hat{a}_k$ where $\hat{a}_k \times \hat{a}_k = 0$

Then "phasor" expression of this general wave is:

$$\vec{H}(x, y, z) = \vec{H}_0 e^{-jk_x x - jk_y y - jk_z z}$$

where k_x, k_y, k_z are wave numbers in directions $\hat{x}, \hat{y}, \hat{z}$. If we define the wave number vector $\vec{k} = k_x \hat{a}_x + k_y \hat{a}_y + k_z \hat{a}_z$ and noting that radius vector $\vec{R} = x \hat{a}_x + y \hat{a}_y + z \hat{a}_z$

$$\vec{H}(\vec{R}) = \vec{H}_0 e^{-j[(k_x \hat{a}_x + k_y \hat{a}_y + k_z \hat{a}_z)(x \hat{a}_x + y \hat{a}_y + z \hat{a}_z)]} = \vec{H}_0 e^{-j\vec{k} \cdot \vec{R}}$$

scalar function

Remembering $\nabla \times \vec{H} = (j\omega\epsilon) \vec{E}$ for time-harmonic waves:

$$\vec{E}(\vec{R}) = \frac{1}{j\omega\epsilon} \nabla \times \vec{H}(\vec{R}) = \frac{1}{j\omega\epsilon} \nabla \times (\vec{H}_0 e^{-j\vec{k} \cdot \vec{R}})$$

Recall identity $\nabla \times (f\vec{A}) = \nabla f \times \vec{A} + f(\nabla \times \vec{A})$ where f is a scalar function

Since $\vec{H}_0$ is independent of $\vec{R}$, $\nabla \times \vec{H}_0 = 0$, second term above is zero.

$$\begin{aligned} \vec{E}(\vec{R}) &= \frac{1}{j\omega\epsilon} \nabla(e^{-j\vec{k} \cdot \vec{R}}) \times \vec{H}_0 = \left[\frac{-j\vec{k}}{j\omega\epsilon} e^{-j\vec{k} \cdot \vec{R}} \times \vec{H}_0 \right] \\ &= -\frac{e^{-j\vec{k} \cdot \vec{R}}}{\omega\epsilon} (\vec{k} \times \vec{H}_0) \end{aligned}$$

$$\|\hat{a}_k\| = 1$$

$\rightarrow$ Define $k = \|\vec{k}\| = \sqrt{k_x^2 + k_y^2 + k_z^2}$. Then $\vec{k} = k \hat{a}_k \rightarrow$ unit direction vector $\hat{a}_k$

$$\vec{E}(\vec{R}) = -\frac{k}{\omega\epsilon} [\hat{a}_k \times (\vec{H}_0 e^{-j\vec{k} \cdot \vec{R}})] = -\frac{\omega\sqrt{\mu\epsilon}}{\omega\epsilon} [\hat{a}_k \times \vec{H}(\vec{R})] = \boxed{-\eta [\hat{a}_k \times \vec{H}(\vec{R})]}$$

where $\eta = \sqrt{\frac{\mu}{\epsilon}}$ is intrinsic impedance of the medium.

3) Problem 7.4

$$\vec{H} = \hat{a}_z 4 \times 10^{-6} \cos(10^7 \pi t - k_0 y + \frac{\pi}{4}) \quad (A/m)$$

↳ plane wave travelling in air

a) Comparing it with the general plane wave eq: $\vec{H} = \hat{a}_k H_0 \cos(\omega t - \vec{k} \cdot \vec{r} + \theta)$
Wave is travelling at +y direction and it's a TEM wave with $\vec{H}$ in $\hat{a}_z$ direction

$$\omega = 10^7 \pi, \quad k_0 = \omega \sqrt{\mu_0 \epsilon_0} \underset{\text{since medium is air}}{=} 10^7 \pi \sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}} \cong \boxed{0.105 \text{ rad/m}} \\ \boxed{= \pi/30}$$

For $H_z = 0$, we need $10^7 \pi t - k_0 y + \frac{\pi}{4} = \frac{\pi}{2} \pm n\pi$ where $n=0,1,2,\dots$
At $t = 3 \times 10^{-3} s$, we have

$$3\pi \times 10^4 - \frac{\pi}{30} y + \frac{\pi}{4} = \pm n\pi + \frac{\pi}{2}, \quad \frac{\pi}{30} y = (3\pi \times 10^4 - \frac{\pi}{4} \pm n\pi)$$

Since $3\pi \times 10^4$ is equivalent to 0 in radians (integer multiple of 2π)

$$\frac{\pi y}{30} = (\pm n\pi - \frac{\pi}{4}) \Rightarrow \boxed{y = \pm 30n - 7.5 \text{ m}} \quad n=0,1,2,\dots$$

b) In terms of phasors: $\vec{H}_s = \hat{a}_z 4 \times 10^{-6} e^{-j(k_0 y - \frac{\pi}{4})} \quad (\frac{A}{m}) \quad (\text{cosine reference})$

From the last problem we know that $\vec{E}_s = -\eta (\hat{a}_k \times \vec{H}_s)$

Recalling $\eta = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.85 \times 10^{-12}}} \cong 120\pi \cong 337 \Omega$ is the intrinsic impedance of air

$$\vec{E}_s = -120\pi (\underbrace{\hat{a}_y}_{\hat{a}_k: \text{direction of prop}} \times \hat{a}_z) 4 \times 10^{-6} e^{-j(0.105y - \frac{\pi}{4})} = -\hat{a}_x 1.508 \times 10^{-3} e^{-j0.105y + j\frac{\pi}{4}} \quad V/m$$

* Converting it back to the time domain:

$$\boxed{\vec{E}(y,t) = -\hat{a}_x 1.51 \times 10^{-3} \cos(10^7 \pi t - 0.105 y + \frac{\pi}{4}) \quad \frac{V}{m}}$$

4) Problem 7.6

$$\vec{E}(z,t) = \hat{a}_x E_{10} \sin(\omega t - kz) + \hat{a}_y E_{20} \sin(\omega t - kz + \psi)$$

Let the phase of the x component be $\phi = \omega t - kz$

$$\vec{E}(\phi, t) = \hat{a}_x E_{10} \sin(\phi) + \hat{a}_y E_{20} \sin(\phi + \psi)$$

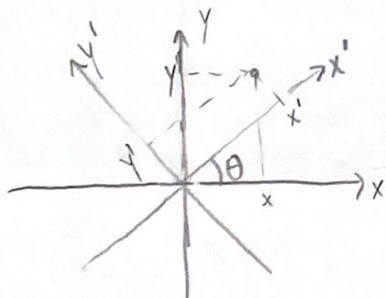
$$E_x = E_{10} \sin \phi, \quad E_y = E_{20} [\sin \phi \cos \psi + \cos \phi \sin \psi]$$

$$\frac{E_x}{E_{10}} = \sin \phi, \quad \frac{E_y}{E_{20}} = \frac{E_x}{E_{10}} \cos \psi + \sqrt{1 - \frac{E_x^2}{E_{10}^2}} \sin \psi$$

$$\left(\frac{E_y}{E_{20}} - \frac{E_x}{E_{10}} \cos \psi \right)^2 = \left(\frac{E_y}{E_{20}} \right)^2 + \left(\frac{E_x}{E_{10}} \right)^2 \cos^2 \psi - \frac{2 E_x E_y \cos \psi}{E_{10} E_{20}} = \left(1 - \frac{E_x^2}{E_{10}^2} \right) \sin^2 \psi$$

$$\left(\frac{E_y}{E_{20}} \right)^2 + \left(\frac{E_x}{E_{10}} \right)^2 - \frac{2 E_x E_y \cos \psi}{E_{10} E_{20}} = \sin^2 \psi$$

This is the equation of a rotated ellipse. To see why let's construct the equations of a rotated ellipse



A point (x, y) will have coordinates (x', y') in the coordinate system rotated by θ degrees:

$$\begin{aligned} x &= x' \cos \theta - y' \sin \theta & x' &= x \cos \theta + y \sin \theta \\ y &= x' \sin \theta + y' \cos \theta & y' &= x \sin \theta - y \cos \theta \end{aligned}$$

Assume that there exist an ellipse whose major and minor axis on x' and y' axes. Then equation of the ellipse in the rotated coordinate axes is:

$$\left(\frac{x'}{a} \right)^2 + \left(\frac{y'}{b} \right)^2 = 1 \quad \text{or} \quad b^2 x'^2 + a^2 y'^2 = a^2 b^2$$

Replacing x' and y' with their correspondants :

$$b^2 x^2 \cos^2 \theta + b^2 y^2 \sin^2 \theta + b^2 2xy \cos \theta \sin \theta + a^2 x^2 \sin^2 \theta + a^2 y^2 \cos^2 \theta - a^2 2xy \cos \theta \sin \theta = a^2 b^2$$

$$x^2 (b^2 \cos^2 \theta + a^2 \sin^2 \theta) + y^2 (b^2 \sin^2 \theta + a^2 \cos^2 \theta) - 2xy \cos \theta \sin \theta (a^2 - b^2) = a^2 b^2$$

$$x^2 \left[\underbrace{\left(\frac{\cos \theta}{a} \right)^2 + \left(\frac{\sin \theta}{b} \right)^2}_A \right] + y^2 \left[\underbrace{\left(\frac{\sin \theta}{a} \right)^2 + \left(\frac{\cos \theta}{b} \right)^2}_B \right] - 2xy \cos \theta \sin \theta \underbrace{\frac{(a^2 - b^2)}{a^2 b^2}}_C = 1$$

Recalling our wave eq:

$$E_x^2 \frac{1}{E_{10}^2 \sin^2 \psi} + E_y^2 \frac{1}{E_{20}^2 \sin^2 \psi} - 2 E_x E_y \frac{\cos \psi}{E_{10} E_{20} \sin^2 \psi} = 1$$

Notice that $\frac{2C}{B-A} = \frac{\sin 2\theta (a^2 - b^2)}{a^2 b^2 \cos 2\theta \left(\frac{1}{b^2} - \frac{1}{a^2} \right)} = \tan 2\theta$

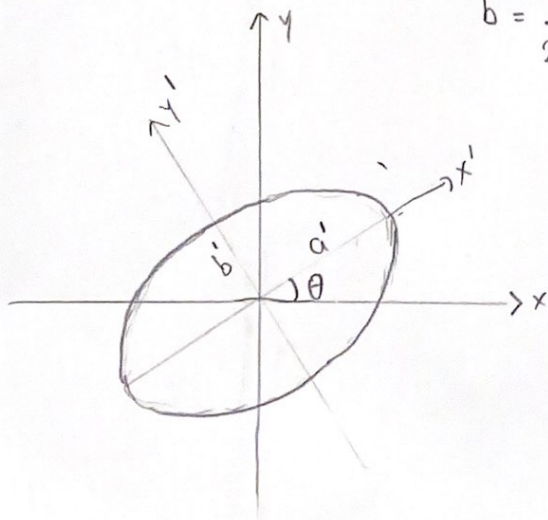
Then we find $\tan 2\theta = \frac{2 \cos \psi}{E_{10} E_{20} \left(\frac{1}{E_{10}^2} - \frac{1}{E_{20}^2} \right)} = \frac{2 E_{10} E_{20} \cos \psi}{E_{20}^2 - E_{10}^2}$

Angle of rotation: $\theta = \frac{1}{2} \tan^{-1} \left(\frac{2 E_{10} E_{20} \cos \psi}{E_{20}^2 - E_{10}^2} \right)$

→ After some algebra major and minor axes lengths are calculated as

$$a' = \frac{1}{2 \sin^2 \psi} \left(\frac{1}{E_{10}^2} + \frac{1}{E_{20}^2} \right) + \frac{1}{\sin^2 \psi} \sqrt{\frac{1}{4} \left(\frac{1}{E_{10}^2} - \frac{1}{E_{20}^2} \right)^2 + \left(\frac{\cos \psi}{E_{10} E_{20}} \right)^2}$$

$$b' = \frac{1}{2 \sin^2 \psi} \left(\frac{1}{E_{10}^2} + \frac{1}{E_{20}^2} \right) - \frac{1}{\sin^2 \psi} \sqrt{\frac{1}{4} \left(\frac{1}{E_{10}^2} - \frac{1}{E_{20}^2} \right)^2 + \left(\frac{\cos \psi}{E_{10} E_{20}} \right)^2}$$



→ Therefore given wave represents a elliptically polarized wave with the ellipse rotated θ degrees from x-axis and major & minor axes length calculated above